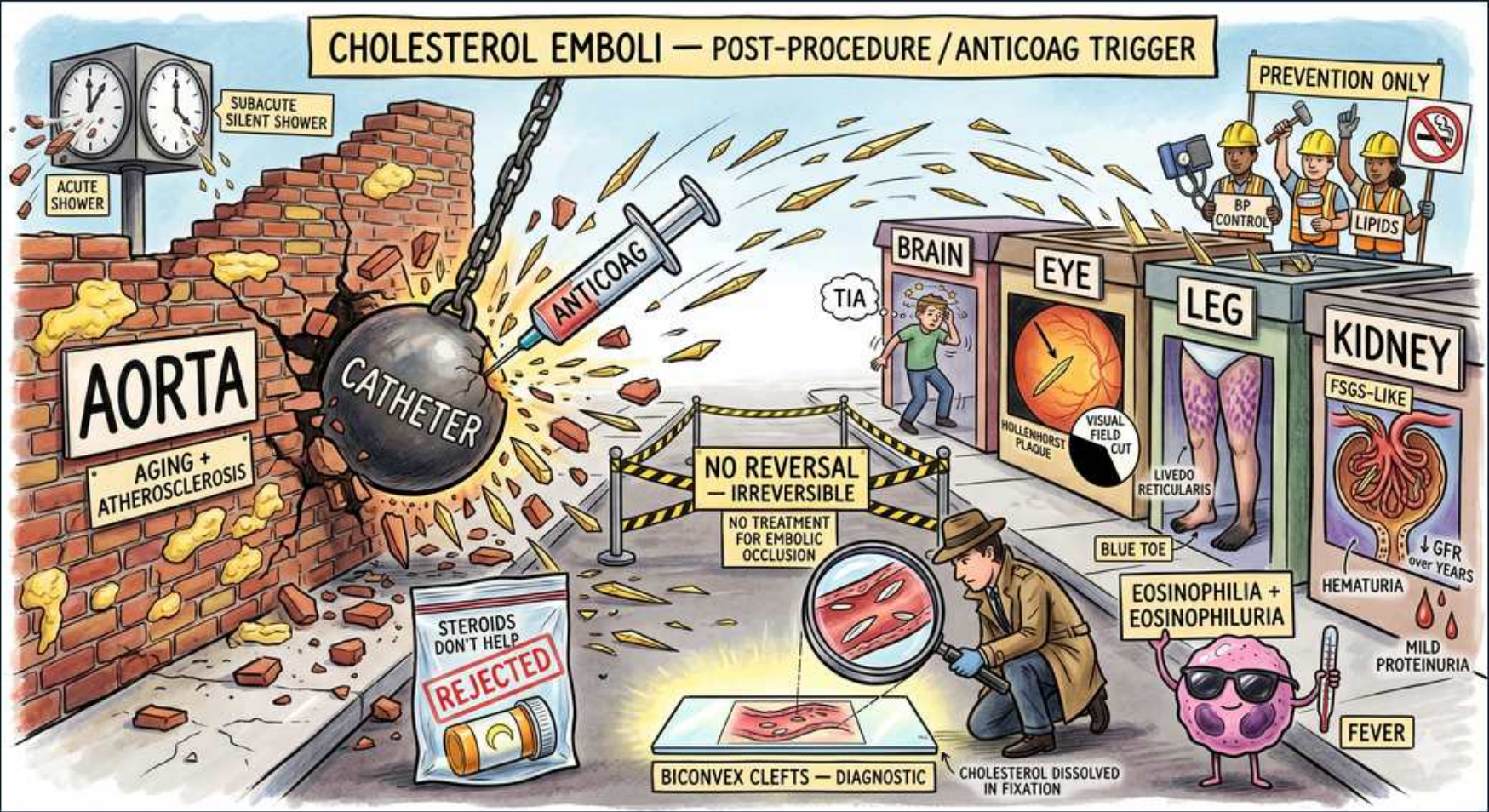


Cholesterol Emboli — Atheroembolic Renal Disease



1. Post-procedure / anticoag trigger

WHAT YOU SEE

Banner: post-procedure / anticoagulation trigger

WHAT IT ENCODES

Atheroembolic (cholesterol crystal embolism) disease is typically triggered by disruption of aortic plaque — recent angiography/catheterization, vascular surgery, or anticoagulation/thrombolysis. Showers of cholesterol crystals lodge in small arteries of kidney, skin, and gut.

BOARD TRAP

Suspect cholesterol emboli (not contrast nephropathy) when AKI appears DAYS-TO-WEEKS after a procedure with skin/eye findings — contrast nephropathy peaks within 3–5 days and resolves.

2. Aorta — aging atherosclerosis

WHAT YOU SEE

Aorta crumbling: aging + atherosclerosis source

WHAT IT ENCODES

The embolic source is an atherosclerotic, often aneurysmal aorta in an elderly arteriopath. Cholesterol crystals released from ulcerated plaques shower distally into end-arteries.

BOARD TRAP

The source is the AORTA's atheromatous plaque — this is an embolic (not thrombotic) and not an immune/glomerular primary process.

3. Catheter trigger

WHAT YOU SEE

Wrecking ball catheter striking plaque

WHAT IT ENCODES

Mechanical disruption by an intra-arterial catheter (cardiac cath, angiography, aortic surgery) is the classic precipitant, dislodging crystals. Onset of renal dysfunction is often subacute, days to weeks afterward.

BOARD TRAP

A subacute, stepwise rising creatinine WEEKS after catheterization favors cholesterol emboli over the early, transient creatinine bump of contrast-induced nephropathy.

4. Anticoagulation trigger

WHAT YOU SEE

Syringe: anticoagulation as a trigger

WHAT IT ENCODES

Anticoagulation and thrombolytic therapy can precipitate cholesterol emboli by preventing protective clot over ulcerated plaque, allowing crystal release. Warfarin is a classic offender.

BOARD TRAP

Anticoagulation can CAUSE cholesterol embolization — bleeding into/destabilizing plaque — so 'more anticoagulation' is not the fix.

5. Acute vs subacute shower

WHAT YOU SEE

Clocks: acute vs subacute silent shower

WHAT IT ENCODES

Embolization can be acute (a single large shower) or recurrent/subacute ('silent' stuttering showers) producing a stepwise decline in renal function over weeks. This tempo distinguishes it from the rapid course of contrast or ATN.

BOARD TRAP

The classic STEPWISE/subacute creatinine rise over weeks separates atheroemboli from the abrupt-and-recovering pattern of contrast nephropathy or ATN.

6. Biconvex clefts — diagnostic

WHAT YOU SEE

Biopsy: biconvex needle-shaped clefts, cholesterol dissolved in fixation

WHAT IT ENCODES

Biopsy shows pathognomonic biconvex, needle-shaped CLEFTS within small arteries/arterioles — the cholesterol itself dissolves during tissue processing, leaving the empty cleft. This confirms the diagnosis.

BOARD TRAP

The cholesterol crystals DISSOLVE in fixation; you see empty BICONVEX CLEFTS, not visible cholesterol — recognizing the cleft is the diagnostic clue.

7. Eosinophilia + eosinophiluria

WHAT YOU SEE

Eosinophilia + eosinophiluria, fever

WHAT IT ENCODES

A characteristic lab triad includes transient peripheral EOSINOPHILIA, eosinophiluria, and hypocomplementemia, sometimes with fever and elevated inflammatory markers — a systemic embolic syndrome mimicking vasculitis or AIN.

BOARD TRAP

Eosinophilia + AKI also occurs in acute interstitial nephritis — but cholesterol emboli add livedo reticularis, blue toes, and a recent vascular procedure.

8. Blue toe / livedo reticularis

WHAT YOU SEE

Leg panel: livedo reticularis + blue toe syndrome

WHAT IT ENCODES

Peripheral skin findings — livedo reticularis and 'blue toe syndrome' with intact distal pulses — are key clinical clues that crystals are showering systemically. Look at the feet in any post-cath AKI.

BOARD TRAP

Blue toes WITH palpable pedal pulses is classic for cholesterol emboli (microvascular), distinguishing it from large-vessel limb ischemia (absent pulses).

9. Ocular Hollenhorst plaques

WHAT YOU SEE

Eye panel: cholesterol plaque in retinal artery (Hollenhorst), visual field

WHAT IT ENCODES

Bright refractile cholesterol crystals at retinal artery bifurcations (Hollenhorst plaques) confirm systemic atheroembolization and may cause amaurosis/visual field loss. Brain showers can cause TIA/stroke.

BOARD TRAP

Hollenhorst plaques on fundoscopy are visible evidence of cholesterol emboli — a window into the same process injuring the kidney.

10. Renal — FSGS-like, hematuria

WHAT YOU SEE

Kidney: FSGS-like, falling GFR over weeks, hematuria, mild proteinuria

WHAT IT ENCODES

Renally, cholesterol emboli cause a subacute decline in GFR with mild proteinuria and bland-to-mild hematuria; chronic ischemia can produce secondary FSGS-like glomerular changes. Prognosis for renal recovery is often poor.

BOARD TRAP

Proteinuria is usually MILD (not nephrotic) and the picture is ischemic — not a primary nephrotic glomerulopathy despite any FSGS-like change.

11. No reversal — irreversible

WHAT YOU SEE

Sign: no reversal, irreversible, no treatment for embolic occlusion

WHAT IT ENCODES

There is no specific therapy to reverse established crystal embolization — management is supportive (statins, blood pressure control, avoiding further instrumentation/anticoagulation) and treating organ failure. Outcomes are frequently poor.

BOARD TRAP

Unlike a thrombus, cholesterol emboli have NO thrombolytic/reversal therapy — anticoagulation may worsen it.

12. Steroids don't help / rejected

WHAT YOU SEE

Rejected gavel: steroids don't help

WHAT IT ENCODES

Despite the eosinophilia and vasculitis-like presentation, corticosteroids are NOT proven beneficial for atheroembolic renal disease. Management focuses on risk-factor control and statins.

BOARD TRAP

Don't reflexively give steroids for the eosinophilia — they are ineffective here, unlike in true AIN or vasculitis.

13. Prevention only

WHAT YOU SEE

Prevention: BP control, lipids, statins, avoid instrumentation

WHAT IT ENCODES

Because there is no cure, prevention dominates: aggressive statin/lipid lowering, blood pressure control, smoking cessation, and minimizing unnecessary arterial instrumentation or anticoagulation in high-risk arteriopathies.

BOARD TRAP

The 'treatment' of cholesterol emboli is really PREVENTION and risk-factor modification — once crystals shower, the renal injury is largely fixed.
